

Urban Heat Islands: Causes, Health Impacts and Mitigation

Policy brief prepared for the City Sustainability Committee

Summary

Cities are often 1 to 3 degrees C warmer than surrounding rural areas during the day and up to 10 degrees C warmer at night. This brief explains why and recommends actions.

Causes

- Dark asphalt and roofs absorb sunlight and release heat after sunset.
- Few trees and little vegetation means less cooling from evapotranspiration.
- Tall buildings trap heat and block wind in narrow street canyons.
- Waste heat from air conditioners, vehicles and industry.

Health impacts

Heat is the deadliest weather hazard in many countries. Elderly residents, outdoor workers and people without air conditioning are most at risk. Neighborhoods with the least tree canopy are often lower-income and measurably hotter.

Mitigation options

1. Cool roofs: reflective coatings lower roof surface temperature by 20 to 30 degrees C.
2. Street trees: target 30 percent canopy cover, prioritizing the hottest districts.
3. Cool pavements and permeable surfaces in parking lots.
4. Green roofs on public buildings.
5. Cooling centers and heat warning systems for vulnerable residents.

Recommendation

Fund a tree-planting program in the five hottest census tracts and require reflective roofing on all new commercial buildings starting next year.